MODULE 07 · SECTION 1 - FUNDAMENTALS OF GEN AI

The gen AI landscape in five layers

Infrastructure, models, platforms, agents and applications, and what each one means for a leader.

SECTION 1 of 4 EXAM WEIGHT ~30% MODULE 7 of 33

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Every gen AI product sits on a stack. Knowing the layer tells you **who builds it, who buys it, and who runs it.**

The stack, from the ground up

01
Infrastructure
Compute and storage to train and run models.

02
Models
Foundation models that understand and generate.

03
Platforms
Tools to build, deploy and manage AI solutions.

04
Agents
Systems that reason, plan and use tools to act.

05
Applications
What people actually use, day to day.

ON THE EXAM

These five names are the exam guide's own. Learn them in this order, bottom to top.

The stack as a restaurant

LAYER	IN A RESTAURANT
Infrastructure	The building, the ovens, the power and the cold store
Models	The chef, with years of training and skill
Platforms	The kitchen system: stations, recipes, ordering, hygiene checks
Agents	The waiter who takes your order, checks stock, and acts
Applications	The dining room and the menu: all the guest ever sees

THE TAKEAWAY

Guests judge the meal, not the ovens. But without the lower layers, nothing reaches the table.

Infrastructure: the power underneath

- **Compute** to train and run models: TPUs and GPUs.
- **TPUs** are Google's own chips, designed for AI work.
- **Storage and networking** for training data and fast connections.
- **AI Hypercomputer** is Google's name for the integrated system of chips, storage and software.
- **Rarely bought directly** by business teams; it sits under everything else.

> ON THE EXAM
>
> Computing power and storage for training data: that is the infrastructure layer, every time.

Models: the intelligence

- **Foundation models** that understand and generate text, images, audio, video and code.
- **Google's own:** Gemini, Gemma, Imagen and Veo, covered in module 8.
- **Partner and open models** too, through Model Garden.
- **Pre-trained** by the model maker, so you start with the hard part done.
- **A model alone** is not a product. It needs the layers above to be useful.

> ON THE EXAM
>
> Access to pre-trained models that can be adapted: that is the models layer.

Platforms: where solutions are built

- **Tools to build, deploy and manage** AI solutions on top of models.
- **Agent Platform**, formerly Vertex AI: Model Garden, Agent Studio, tuning, evaluation, MLOps.
- **Google AI Studio** for fast experiments with Gemini models.
- **From low-code to code-first,** so different skill levels can build.
- **Not** the end-user interface. Customers rarely see the platform.

> ON THE EXAM
>
> Tools to build, deploy and manage AI solutions: that is the platform layer.

Agents: from answering to acting

- **Agents** pursue a goal: they reason, plan, and use tools to take action.
- **They act,** not just answer: check an order, book a collection, update a record.
- **Examples:** agents in Gemini Enterprise, contact-centre agents, custom agents built with ADK.
- **The fastest-moving layer** in 2026, and the newest in the guide.
- **More autonomy** means more need for limits, permissions and oversight.

> ON THE EXAM
>
> Uses tools and takes action toward a goal, with little step-by-step instruction: that is the agents layer.

Applications: what people use

- **The interface** where people interact with AI, often without thinking about it.
- **Google's:** the Gemini app, Gemini for Google Workspace, Gemini Notebook, the Gemini Enterprise app.
- **Yours:** the chat on your website, the assistant inside your own product.
- **Fastest to value:** buy, switch on, train people, measure.
- **Least customisable:** you get what the application offers.

> ON THE EXAM
>
> Lets users interact with and use AI capabilities directly: that is the applications layer.

Google's offerings, layer by layer

LAYER	GOOGLE EXAMPLES
Applications	Gemini app, Gemini for Google Workspace, Gemini Notebook, Gemini Enterprise app
Agents	Gemini Enterprise agents, Customer Engagement Suite agents, ADK-built agents
Platforms	Agent Platform (formerly Vertex AI), Google AI Studio
Models	Gemini, Gemma, Imagen, Veo, plus partner models in Model Garden
Infrastructure	TPUs, GPUs, AI Hypercomputer, Google's global data centres

ON THE EXAM

Google offers something at every layer. That full stack is one of the strengths the exam asks about.

Lower layers or upper layers?

WORK LOWER IN THE STACK

More control

- Choose and adapt your own model
- Build exactly the solution you want
- Needs skills, time and budget
- You own more of the running

WORK HIGHER IN THE STACK

Faster value

- Adopt a ready-made agent or app
- Results in days, not months
- Few technical skills needed
- Less room to customise

THE TIEBREAKER

Go only as low as the problem demands. Speed and missing skills push you up; unique needs push you down.

Why the layers matter to a leader

- **Buying decisions:** each layer is a different purchase and a different supplier conversation.
- **Skills:** lower layers need engineers; upper layers need change management.
- **Differentiation** rarely comes from the chips. It comes from your data and your workflows.
- **Openness:** choosing open models and standards at each layer limits lock-in.
- **Responsibility** is shared: the provider runs the lower layers, you own the use.

THE TAKEAWAY

Your advantage lives in the top layers and in your own data, not in the infrastructure.

Wrenfield's projects, by layer

WRENFIELD PROJECT	LAYER IT WORKS AT
Marketing drafts copy with Gemini in Google Docs	Applications
Staff ask policy questions in Gemini Enterprise	Applications, with agents
A returns agent checks orders and books collections	Agents
Developers build a catalogue-copy tool for the Dutch site	Platforms and models
Chips, storage and data centres	Infrastructure, run by Google

THE TAKEAWAY

Wrenfield works at four layers and buys the fifth. That is typical for a business of its size.

> A board member has read that Google designs its own AI chips and uses them to train models such as Gemini.
>
> She asks which layer of the generative AI landscape that work belongs to.

Which layer is it?

A Applications

B Infrastructure

C Agents

D Models

◆ PAUSE AND DECIDE

Which layer is it?

A Applications

What users touch. Chips sit at the bottom.

B Infrastructure

Chips and compute to train and run models.

C Agents

Agents use tools to act. They are not hardware.

D Models

Gemini is a model, but the chips that train it are infrastructure.

THE SIGNAL

Chips, compute and storage for training: infrastructure.

> Wrenfield's developers want one place to choose a foundation model, test prompts and tune it on brand copy. They also want to evaluate it and deploy it behind their own app.
>
> They will manage versions and monitor it there too.

Which layer are they mainly working in?

A Applications

B Infrastructure

C Platforms

D Models

◆ PAUSE AND DECIDE

Which layer are they mainly working in?

A Applications

Their finished app will be one, but they are building it.

B Infrastructure

Google runs the chips. The team never touches them.

C Platforms

Tools to build, deploy and manage, in one place: Agent Platform.

D Models

They choose and tune a model, but the work happens on the platform.

THE SIGNAL

Choose, tune, evaluate, deploy, manage: a list of platform jobs.

> Today, Wrenfield's staff ask the Gemini app to draft replies to returns emails, then process each return by hand.
>
> A new system will read the email, check the order, book a courier collection and confirm with the customer, with no person in between.

Which layer does the new capability add?

A Models

B Applications

C Agents

D Infrastructure

◆ PAUSE AND DECIDE

Which layer does the new capability add?

A **Models**
A model is still underneath, but it is not the new part.

B **Applications**
Drafting in the Gemini app is today's application, not the change.

C **Agents**
Uses tools to check, book and confirm: acting, not just answering.

D **Infrastructure**
Unchanged, and run by Google.

THE SIGNAL

From drafting a reply to doing the work with tools: that is the jump to agents.

> A 40-person charity with no technical staff wants help summarising reports and drafting letters to funders. Everyone already uses Google Workspace.
>
> The trustees ask where in the stack they should invest.

What should the charity do?

A Rent TPUs and train a small model of its own

B Adopt an application such as Gemini for Google Workspace

C Fine-tune an open model on its past letters

D Build a custom solution on Agent Platform

◆ PAUSE AND DECIDE

What should the charity do?

A Rent TPUs and train a small model of its own

Infrastructure work, far beyond its skills and budget.

B Adopt an application such as Gemini for Google Workspace

Top layer: ready-made, in tools they already use.

C Fine-tune an open model on its past letters

Model work that needs skills the charity lacks.

D Build a custom solution on Agent Platform

Platform work needs developers. Nothing custom is needed.

THE SIGNAL

Common tasks, no technical staff, already on Workspace: buy at the top of the stack.

Module 7 in five lines

- Five layers, bottom up: infrastructure, models, platforms, agents, applications.
- Infrastructure is compute and storage; platforms are tools to build, deploy and manage.
- Agents act with tools toward a goal; applications are what people use.
- Google offers something at every layer, from TPUs to the Gemini app.
- Go only as low in the stack as the problem demands.